

Gabriel Melgarejo

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EDUCATION

Foothill College

3.70/4.00 GPA, Mechanical Engineering transfer program, Emphasis in Robotics

Expected June 2027

Manhattanville University

3.82/4.00 GPA, Undeclared Major

September 2021- May 2023

WORK EXPERIENCE

Undergraduate Researcher, AI & Robotics Lab (ETI), Foothill College

January 2026-Present 2025

- Developed a computer vision system using an Azure Kinect DK camera, OpenCV, and Python to detect and classify 2D shapes in real time, building foundational skills in robotic perception.
- Implemented a complete computer vision pipeline including Gaussian blur noise reduction, Canny edge detection, contour tracing, and geometric classification using circularity formula and vertex counting.
- Applied linear algebra concepts including dot products for grayscale conversion, image moment calculations for centroid detection, and matrix operations for real-time frame processing at 30fps.
- Extended system to 3D shape classification by fusing RGB contour detection with Azure Kinect depth data to distinguish flat vs volumetric objects and estimate real-world distance in meters.

Undergraduate Research Intern, Winetraub Lab, Stanford University

July 2025- September 2025

- Designed and prototyped a sample holder for optimizing OCT imaging through multiple design iterations, using CAD modeling and 3D printing.
- Developed a final design integrating an O-ring seal and snap-fit locking mechanism, achieving water-tight containment, constant compression, and reliable alignment. This led to filing a patent with Stanford University.
- Validated the design through prototype testing, ensuring durability, leak-free performance, and ease of assembly/disassembly.
- My device will contribute to the transition of the non-invasive biopsy method used in my lab to Stanford Hospital, introducing OCT sample preparation and aid pathologists in early cancer detection.
- Applied problem-solving, collaboration, and innovation to deliver a design that can be adapted for future biomedical and optical devices.

Engraver, Crown Awards Inc., United States

January 2023- May 2023

- Operated laser and rotary engraving machines to personalize trophies, plaques, and medals.
- Read and interpret customer orders to ensure accurate text, fonts, and logo placement.
- Conducted quality control checks to verify precision, alignment, and overall engraving quality.
- Prepared and positioned materials for engraving, including cleaning and polishing surfaces.
- Performed basic machine maintenance, such as cleaning and adjusting engraving settings.
- Collaborated with the production team to meet order deadlines efficiently.

TECHNICAL SKILLS & RELEVANT COURSEWORK

- **Software:** SolidWorks, Onshape, Fusion, Microsoft Office
- **Programming:** Python, Java, Arduino Uno, MATLAB
- **Manufacturing:** 3D Printing (Bambu Lab), Soldering, Sandblasting, Embroidery, Laser Cut and Engraver (BossLaser Ls-3655)
- **Relevant Coursework:** Engineering Graphics; Calculus 1, 2,3; Linear algebra; Physics Kinematics, Electromagnetism
- **Languages:** English, Spanish, Italian